

Conformal-Coat — Standard Operating Procedure

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- 1. Purpose
- 2. Normal operating envelope
- 3. Idle reasons
- 4. Faults & corrective actions
- 5. Preventive maintenance schedule
- 6. References

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Audience: Maintenance Tech · Line Manager · Quality Worker
Applies to: Line-E, station 8, Conformal-Coat

1. Purpose

Conformal-Coat applies a thin protective polymer film over the tested electronics module to shield it from moisture, contamination, and vibration. UV cure follows the coating step. This is the final station of Line-E.

2. Normal operating envelope

- Ideal cycle time: **25 s**
- Max acceptable cycle time: **32 s**
- Expected reject rate: **≤ 2 %**
- Fault probability per cycle: **0.4 %**
- Input buffer capacity: **5 parts**

3. Idle reasons

idle_reason	Meaning here	First check
Starved	Functional-Test (station 7) is not feeding boards.	Verify Functional-Test state.
Blocked	Outbound conveyor full.	Verify part-out.

4. Faults & corrective actions

4.1 Nozzle-Clog

- **Symptoms in telemetry:** fault_type = "Nozzle-Clog". Coating pattern fails; thickness rejects rise.
- **Likely root cause:** dried coating in the nozzle from idle period.
- **Corrective action:**
 - 1. Flush with solvent.

2. Clean nozzle in ultrasonic if needed.
- **Restart criteria:** spray pattern verified against the test card.

4.2 Viscosity-Off

- **Symptoms in telemetry:** `fault_type = "Viscosity-Off"`. Thickness highly variable; downstream inspection rejects clustering.
- **Likely root cause:** coating batch out of spec, or solvent evaporation.
- **Corrective action:**
 1. Sample coating; measure viscosity.
 2. Adjust thinner to recipe.
- **Restart criteria:** viscosity within recipe range.

4.3 UV-Lamp-Fail

- **Symptoms in telemetry:** `fault_type = "UV-Lamp-Fail"`. Cure incomplete; downstream stickiness rejects rise.
- **Likely root cause:** UV lamp at end of life, or driver failure.
- **Corrective action:**
 1. LOTO. Replace UV lamp.
 2. Verify UV intensity with radiometer.
- **Restart criteria:** UV dose within recipe range.

5. Preventive maintenance schedule

- **Daily:** purge gun; UV intensity check.
- **Weekly:** clean nozzle and lamps.
- **Monthly:** replace UV lamp at scheduled hours.
- **Quarterly:** atomization-system tune.

6. References

- Maintenance_Workflow_Overview.pdf
- Safety_Lockout_Tagout_Procedure.pdf
- Line-E_07_Functional-Test_SOP.pdf
- Production_Lines_Reference.pdf